

Taguchi's Quality Loss Function (QLF) & Robust Design

PREPARED BY

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TARGET INDUSTRY

Consumer Electronics &
Hardware

COMPONENT / FEATURE

Smartwatch Unibody Groove
Depth

INDUSTRY BENCHMARKS

Apple Hardware / Samsung
MX

1. Executive Summary & Context

In consumer electronics hardware manufacturing (for top brands like **Apple Hardware Supply Division** and **Samsung MX Operations**), traditional quality checks follow a simple rule: if a dimension is inside the Upper Specification Limit (USL) and Lower Specification Limit (LSL), it passes with zero loss.

In this study, I used **Genichi Taguchi's Quality Loss Function (QLF)** to check what happens when dimensions deviate from the exact target depth (2.00 mm). My goal was to see how small variations inside the tolerance band still create real financial losses because of seal wear, loose fitment, and future warranty risks.

MATHEMATICAL BASELINE & FORMULA SETUP

Quality Loss Formula: $L(y) = k \times (y - m)^2$ | Loss Coefficient: $k = A / (\Delta)^2$

Where: m = Target Value (2.00 mm), Δ = Tolerance Limit (± 0.05 mm), A = Full Rejection / Scrap Cost (₹50,000)

$k = 50,000 / (0.05)^2 = 50,000 / 0.0025 = 20,000,000$ (2 Crore)

2. Data & Calculation Analysis (Traditional QA vs. Taguchi Model)

SAMPLE ID	MEASURED SIZE (Y)	DEVIATION (Y - M)	TRADITIONAL QA	TAGUCHI FINANCIAL LOSS L(Y)	PRACTICAL OPERATIONAL IMPACT
Sample 1	2.00 mm	0.00 mm	PASS (₹0)	₹0	Exact Target Match (Zero Defect)
Sample 2	2.02 mm	+0.02 mm	PASS (₹0)	₹8,000	Hidden Loss (Minor Sealing Degradation)
Sample 3	2.05 mm (USL)	+0.05 mm	PASS (₹0)	₹50,000	Borderline Limit (Max Tolerance Loss)
Sample 4	2.07 mm (Out)	+0.07 mm	REJECT (₹50k)	₹98,000	Critical Defect (Assembly Failure)

3. Engineering Insights & Key Takeaways

1. Sample 1 as the Perfect Target Standard

Sample 1 (2.00 mm) hits the exact mathematical target size. Since it does not drift toward USL or LSL limits, it generates zero loss. Staying right on target reduces internal failure costs (rework/scrap) and external failure costs (warranty returns). In fact, appraisal costs also go down because we don't need excessive end-of-line sorting.

2. Catching Hidden Financial Losses (Samples 2 & 3)

Traditional inspection calls Sample 2 (2.02 mm) and Sample 3 (2.05 mm) fully acceptable. But applying Taguchi's model reveals hidden financial losses of ₹8,000 and ₹50,000. For high-precision suppliers like **Foxconn Technology Group** or **L&T Precision Engineering**, these small dimensional shifts compromise water resistance (IP68) over time and cause field failures.

3. Building a Zero-Defect System with Kaizen & Deming Cycle (PDCA)

To keep parts consistently close to the 2.00 mm target size, quality engineering needs to combine **Kaizen** and the **Deming Wheel (PDCA Cycle)**:

- **Plan:** Identify root causes making dimensions shift away from 2.00 mm (like CNC tool wear or thermal variation).
- **Do & Check:** Run micro-Kaizen fixes on fixture holding and coolant temperature control.
- **Act:** Standardize control limits around the exact target size to keep overall process variation low.

DISCLAIMER & CONTEXT OF WORK

- **Self-Learning & Domain Knowledge Initiative:** This report is created as part of my self-directed learning journey for the **100-Day Quality Engineering Challenge**.
- **Raw / Dummy Data Utilization:** All dimensions, parameters, formulas, and financial values used here are raw, synthetic dummy data prepared strictly for learning and demonstration purposes. No proprietary company data was used.
- **Target Application Alignment:** The framework and analysis shown here are designed to address practical quality engineering problems in modern consumer electronics and precision hardware manufacturing.